Lucas Neumann

Senior Software Engineer — Pricing, Risk & AI Tooling

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Summary

Senior backend engineer (Python, Go) building data pipelines, cloud infrastructure and AI-agent tooling for financial systems. Five years in buy- and sell-side finance: real-time market-data and pricing services, a central-bank FX-rate feed, high-throughput ETL integrations with vendor platforms and large migrations of legacy codebases.

Experience

BWGI · Senior Software Engineer — 2025–present

- Integrated Bloomberg (MARS, Data License), a vendor PMS and other data providers through ETL pipelines covering multiple funds and thousands of instruments daily, reconciled automatically against legacy pricing.
- Designed a new risk data pipeline (risk engine → PostgreSQL → BI) with Greeks and risk-factor serving tables and a data-quality gate (cross-layer reconciliation, robust z-score outliers), run daily per fund.
- Built AI-agent tooling for the risk and pricing teams: a Go toolkit of CLIs and **MCP** servers plus **20+ agent skills** that give LLM agents governed access to internal data and systems, shipped as versioned releases and adopted by the risk team.
- Led the migration off a legacy monorepo, coordinating library and API changes across teams: orchestration moved to Airflow 3 on Kubernetes and the core data stack upgraded across the shared packages, with pricing and risk services kept running in production.
- Primary on-call for production pricing and risk systems.

Bank of America · Software Engineer, AVP — 2023–2025

FICC Emerging Markets. Joined via BRQ (2023); direct hire as AVP (2024).

- Built and ran real-time curve infrastructure for the EM derivatives and FX trading desk: live market data and trader curve marks reaching the sell-side pricing system **within seconds**, plus intraday and EOD curve publication to internal databases; onboarded new engineers to the curve stack.
- Took the real-time **PTAX** integration with Banco Central do Brasil to production and kept it stable — a high-availability service where any failure carries direct reputational risk.
- Developed in one of the industry's largest Python monorepos (15+ years, C++ performance layer): event-driven market-data processing on proprietary NoSQL and messaging systems.

Apex Capital · Software Developer — 2021–2022

Hedge fund

- Cross-area role, working day to day with research, economics and operations: database administration, data analysis and research reports, web scraping for data mining, Streamlit risk dashboards and automation of Excel-based operations workflows.
- Migrated the fund's scheduled jobs from on-premises Task Scheduler to Airflow on AWS, with version control, CI, tests and Docker.

Skills

Engineering
Python, Go, Rust, SQL (PostgreSQL, Snowflake), Airflow/Astronomer, MongoDB, Redis, Docker, Kubernetes, AWS, Datadog, MCP

Financial systems
Bloomberg MARS & Data License, Arcesium, MSCI Barra, S&P Global, Qlik Sense

Domain
Rates, credit, FX, fixed income; derivatives pricing; VaR, expected shortfall, Greeks

Languages
Portuguese (native), English (fluent)

Education

UNICAMP — University of Campinas
MSc, Physical Chemistry
2018–2020
Computational simulation of proteins (CNPq scholarship); daily work with HPC clusters, Python, Julia, C, C++ and Linux.

Projects

git-archaeologist (Rust)
Fast CLI extracting time series from git history (LOC, cohort survival, coupling, hotspots) as TSV, CSV, JSON or Parquet.

ragwatcher (Python)
Incremental local RAG over a directory, served as an MCP tool and CLI.

visidata-rs (Rust)
Terminal data explorer inspired by VisiData: lazily browses CSV, Parquet, JSON, Excel and SQLite; Polars, Ratatui, Lua.

StringArt.jl (Julia)
Generative art: turns photos into string-art patterns.

MonteCarloArt.jl (Julia)
Generative art: pointillist images via Monte Carlo sampling.